# Introduction

## Circular List
## Data Structures and its Applications (UE25CS52A)

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## Quiz : Circular List

1. What is a common way to detect the end of traversal in a circular linked list?

a) Check current == NULL

b) Check current->next_ == NULL

c) Check whether current has returned to the starting node

d) Check whether current->key_ == 0



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## Quiz : Circular List

2. while (temp != NULL) {
```
        // some other code
        temp = temp->next_;
```
}

temp is a pointer to one of the nodes. Comment.

a) Will become infinite loop

b) Will traverse all the nodes and exit the loop

c) Will traverse all but one node and exit the loop

d) Will result in undefined behaviour



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## Quiz : Circular List

3. node_t* temp = ptr_clist->head_; node_t* temp2 = ptr_clist->head_;

```
do {
        temp = temp->next_;
        temp2 = temp2->next_->next_;
} while (temp2 != ptr_clist->head_);
```

Comment.

a) temp2 will always point to a node half way across from the head

b) Both temp and temp2 may be at the head when the loop terminates

c) This may result in an infinite loop if the number of nodes is even

d) None of these



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## Quiz : Circular List

4. Consider a circular list containing only one node.
Which of the following is true?

a) ptr_list->head_ == NULL

b) ptr_list->head_->next_ == NULL

c) ptr_list->head_ == ptr_list->head_->next_

d) ptr_list->head_ may point to any node



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## Quiz : Circular List

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## Quiz : Circular List

5. In a circular linked list with head_ referring a node in the list, what happens when the only node is deleted?

a) head_ remains unchanged

b) head_ becomes NULL

c) head_ ->next_ becomes NULL

d) A new node must be created



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## Quiz : Circular List

5. In a circular linked list with head_ referring a node in the list, what happens when the only node is deleted?

a) head_ remains unchanged

b) head_ becomes NULL

c) head_ ->next_ becomes NULL

d) A new node must be created



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## Circular List : concept

- This is a list
- The last node points to the first
    - Have a pointer to one of the nodes
    - first and last are not very meaningful
- When the elements form a cycle or need to be processed repeatedly,

  the last element should naturally lead back to the first.



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## Circular List: interface

- Initialize / Deinitialize the list
- Insert anywhere in the list
- Delete anywhere in the list
- Traverse forward
- Search for a designated key



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## Circular List: structure

```
//Same as that of a singly linked list
struct node
{
        int key_;
        struct node *next_;
};
typedef struct node node_t;
struct clist
{
        node_t *head_;
};
typedef struct clist clist_t;
```



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## Circular List: implementation

- Initialize / Deinitialize the list
  - Initialize : Make head_ NULL
  - Deinitialize : Traverse and remove nodes until back to the first node;
    Remove that node; ground the head_.
- Insert after/before the first node
  - Insertion is exactly the same
  - Modify the head_ suitably
- Delete after the first node
- Traverse
  - Prefer a do…while loop
  - Stop when the first node is reached again
- Search for a designated key



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## Circular List: Applications

- Multiplayer Turn Management
- Music Playlist – repeat continuously
- Image Slideshow - repeat continuously
- Token Passing - ring-based algorithms



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## Circular List: Josephus problem : problem statement

**Josephus Problem:**

1. Find the last remaining survivor from n people standing in a circle.
2. Counting begins at a specific person
3. Moves around the circle in a fixed direction
4. In each step, k-1 people are skipped, and the k-th person is removed.
5. The process repeats until only one person is left.
6. Find that person.



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## Circular List: Josephus problem : algorithm

The steps involved in solving this problem are:

1. n people are numbered from 1 to n in order and are placed in a circle
2. Starting from a designated start point, we count to kth person.
3. The kth person is eliminated
4. Counting and elimination steps are repeated until a single person is left
5. Original number of that person is the solution.



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## Circular List: Josephus problem : Example

- **n = 7** people
- **k = 3** → eliminate every 3rd person

| Step | Counting | Eliminated |
|---|---|---|
| 1 | 1 → 2 → 3 | 3 |
| 2 | 4 → 5 → 6 | 6 |
| 3 | 7 → 1 → 2 | 2 |
| 4 | 4 → 5 → 7 | 7 |
| 5 | 1 → 4 → 5 | 5 |
| 6 | 1 → 4 → 1 | 1 |
| | | |

4 is the survivor



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# THANK YOU

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